

Paper

Instructions/template for preparing your NOLTA manuscript

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Key Words: NOLTA, template, nolta.cls, up to six words

1. Introduction

Deep learning has achieved remarkable success in various fields, such as image recognition and natural language processing. The generalization performance of deep neural networks has often been improved by increasing model size, the amount of training data, and computational resources. This tendency is often characterized by scaling laws, which describe how model performance improves as these resources are scaled up.

However, such resource-based scaling is not always applicable in practical situations. In some domains, such as industrial anomaly detection and medical applications, it is difficult to collect a large amount of labeled data, especially for rare or abnormal cases. In addition, some applications must be performed under limited computational resources, memory capacity, or inference time. Therefore, it is important to explore alternative learning paradigms that can improve generalization performance without simply relying on larger models, larger datasets, or larger computational budgets.

In recent years, collaborative learning frameworks using multiple models have attracted attention as one possible approach. These frameworks use the predictions of other models as additional supervision during training, thereby improving generalization performance. Since model collaboration is required only during training, these frameworks can improve model performance without increasing the computational cost at inference time.

A representative example of such frameworks is Mutual learning (ML), in which multiple models are



trained simultaneously while referring to each other’s prediction distributions. Unlike conventional teacher-student approaches, ML does not require a fixed pre-trained teacher model. In ML, each model is trained using a loss function consisting of a standard supervised loss and a KL-divergence-based consistency loss between prediction distributions. Previous studies have shown that ML often achieves better generalization performance than independently trained models. However, it remains unclear how ML improves model performance, particularly how interactions between models during training contribute to this improvement.

In this study, we analyze the generalization process of ML by comparing ML models with individual learning (IL) models. We focus on the double descent phenomenon because it contradicts the conventional bias-variance trade-off and involves non-monotonic changes in test error, which can reveal characteristics of the generalization process. We first examine differences in double descent behavior between ML and IL with respect to the number of model parameters and training epochs. As further analyses, we compare ML and IL in terms of variance, loss landscape geometry, and Hessian-based properties, especially across the regimes of double descent.

2. Related Works

2.1 Mutual Learning

Mutual learning is a training framework in which multiple models are trained simultaneously by referring to each other’s predictions. In individual learning, each model is optimized independently using only the ground-truth labels. In contrast, mutual learning introduces interactions among models during training through their prediction distributions.

In ML, the loss function of each model includes both a supervised loss and a mimicry loss. For two models, let p_1 and p_2 denote the prediction distributions of Model 1 and Model 2, respectively. The objective function of each model consists of a cross-entropy loss and a Kullback–Leibler (KL) divergence term:

$$L_{\Theta_1} = L_{\text{CE}}(y, p_1) + D_{\text{KL}}(p_2 \| p_1), \quad (1)$$

$$L_{\Theta_2} = L_{\text{CE}}(y, p_2) + D_{\text{KL}}(p_1 \| p_2). \quad (2)$$

The cross-entropy term supervises each model using the ground-truth label y , while the KL divergence term encourages the prediction distribution of each model to approach that of the other model.

ML improves generalization performance compared with individual learning across various network architectures and tasks, such as image recognition and person re-identification. Compact models also benefit from mutual learning without using a large pre-trained teacher model. Since each model is trained collaboratively but used independently at inference time, ML can improve the training process without increasing the inference cost of a single model.

The effectiveness of ML has been discussed in terms of posterior entropy and robust minima. Because models are initialized differently, their secondary class probabilities can differ even when they predict the same ground-truth class. Matching these prediction distributions increases posterior entropy and may guide models toward flatter minima with better generalization. However, the relationship between mutual learning and the generalization process under changes in model complexity and training time has not been fully characterized.

2.2 Double Descent

Double descent is a phenomenon in which the test error first decreases, then increases, and finally decreases again as model complexity increases. This behavior differs from the conventional U-shaped generalization curve expected from the classical bias-variance trade-off. Figure X illustrates a typical double descent curve, where the training error continues to decrease while the test error exhibits a non-monotonic pattern. Depending on the source of complexity, double descent can be categorized into parameter-wise and epoch-wise double descent. Parameter-wise double descent occurs as the number of model parameters increases, whereas epoch-wise double descent occurs as the number of training epochs increases.

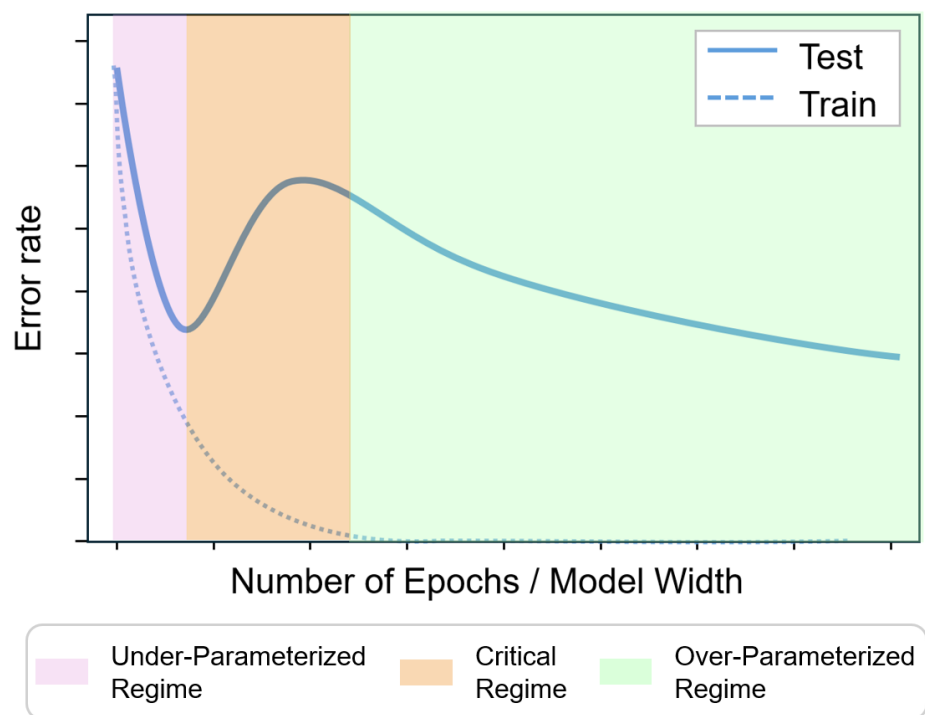


Fig. 1. Caption text.

The curve is often interpreted by dividing the learning process into three regimes, as shown by the shaded regions in Fig. 1: the under-parameterized regime, the critically-parameterized regime, and the over-parameterized regime, shown in purple, orange, and green, respectively. In the under-parameterized regime, the model does not have sufficient capacity to fit the training data, and increasing model complexity tends to reduce the test error. The critically-parameterized regime appears around the interpolation threshold, where the model starts to fit the training data almost perfectly. In this regime, the test error often increases and forms a local maximum, referred to as the peak of double descent. In the over-parameterized regime, the model has sufficient capacity to fit the training data, and further increasing model complexity tends to reduce the test error again.

Previous studies have also investigated factors that affect the peak of double descent. Label noise tends to make the peak more prominent and thus makes the double descent behavior more clearly observable. Data augmentation shifts the peak toward larger models and reduces its height. Ensemble methods have also been reported to reduce the peak.

3. Experiments

4. Discussion

5. Conclusions

Acknowledgments

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